CLUTCH RELEASE BEARING (Continued)

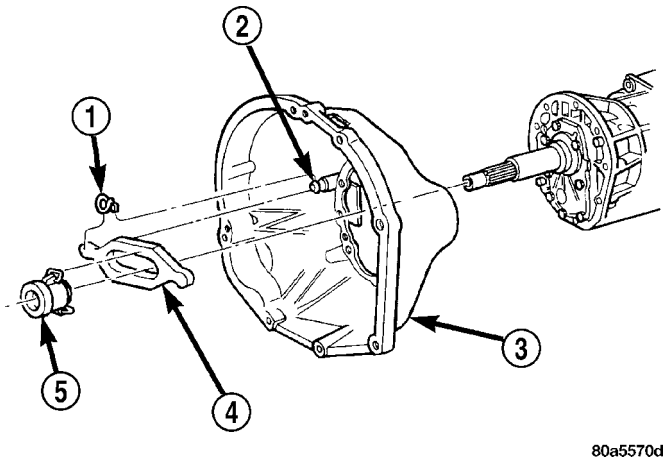

Fig. 3 RELEASE BEARING

1 - RETAINER SPRING
2 - PIVOT BALL STUD
3 - CLUTCH HOUSING
4 - RELEASE FORK
5 - RELEASE BEARING

release fork pivot surface. Lubricate with Mopar high temperature bearing grease or equivalent.

(2) Install new release bearing. Verify bearing is properly secured to release fork.

(3) Install transmission.

FLYWHEEL

DESCRIPTION

STANDARD FLYWHEEL

The standard flywheel is used on the 4.0L engine. The flywheel (Fig. 4) is a heavy plate bolted to the rear of the crankshaft. The flywheel incorporates the ring gear to mesh with the starter. The rear face of the flywheel serves as the driving member to the clutch disc.

DUAL MASS FLYWHEEL

The Dual Mass Flywheel is used on the 2.4L engine (Fig. 5). The flywheel incorporates the ring gear to mesh with the starter. The primary flywheel side is bolted to the crankshaft. The secondary flywheel face serves as the driving member to the clutch disc. Internal springs between the flywheels are use to dampen energy.

OPERATION

The flywheel serves to dampen the engine firing pulses. The heavy weight of the flywheel relative to the rotating mass of the engine components serves to stabilize the flow of power to the remainder of the drivetrain. The crankshaft has the tendency to attempt to speed up and slow down in response to

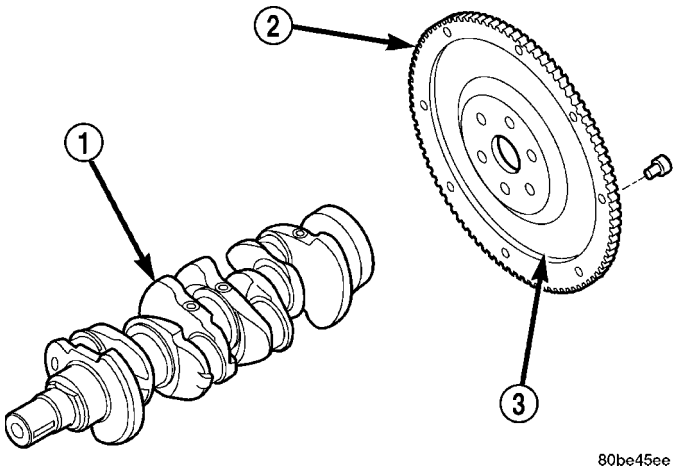

Fig. 4 FLYWHEEL

1 - CRANKSHAFT
2 - RING GEAR
3 - FLYWHEEL

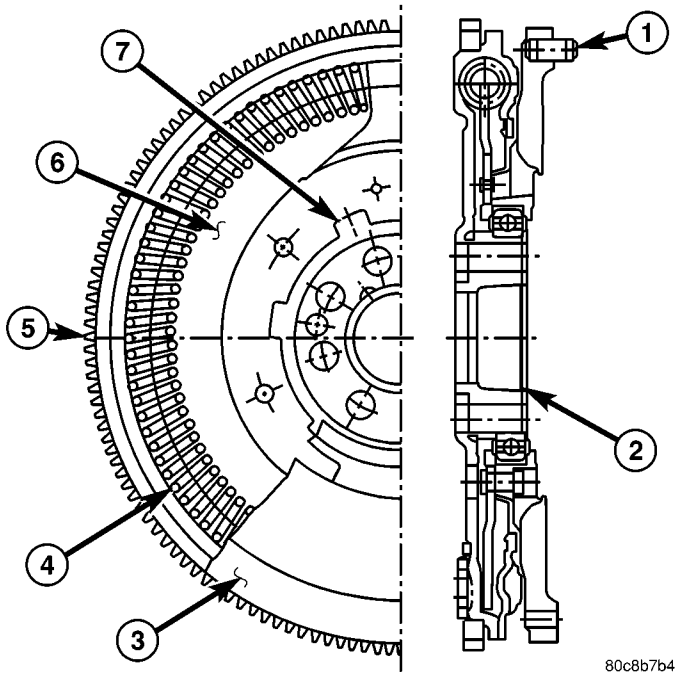

Fig. 5 DUAL MASS FLYWHEEL

1 - LOCATING STUD
2 - BEARING
3 - SECONDARY FLYWHEEL
4 - DAMPER SPRING
5 - RING GEAR
6 - PRIMARY FLYWHEEL
7 - FRICTION DISC

the cylinder firing pulses. The flywheel dampens these impulses by absorbing energy when the crankshaft speeds and releasing the energy back into the system when the crankshaft slows down.

On a Dual Mass Flywheel the additional secondary mass coupled to the transmission lowers the natural frequency of the transmission rotating elements. This